

Method	RGB CNN	Flow CNN	ST-GCN
Accuracy	70.4%	72.8%	72.4%

Table 4: Mean class accuracies on the “Kinetics Motion” subset of the Kinetics dataset. This subset contains 30 action classes in Kinetics which are strongly related to body motions.

also compare our ST-GCN model with the previous state-of-the-art methods on this dataset. Due to the constraint nature of this dataset, we do not use any data augmentation when training ST-GCN models. We follow the standard practice in literature to report cross-subject (X-Sub) and cross-view (X-View) recognition performance in terms of top-1 classification accuracies. The compared methods include Lie Group (?), Hierarchical RNN (?), Deep LSTM (?), Part-Aware LSTM (PA-LSTM) (?), Spatial Temporal LSTM with Trust Gates (ST-LSTM+TS) (?), Temporal Convolutional Neural Networks (Temporal Conv.) (?), and Clips CNN + Multi-task learning (C-CNN+MTLN) (?). Our ST-GCN model, with rather simple architecture and no data augmentation as used in (?; ?), is able to outperform previous state-of-the-art approaches on this dataset.

Discussion. The two datasets in experiments have very different natures. On Kinetics the input is 2D skeletons detected with deep neural networks (?), while on NTU-RGB+D the input is from Kinect depth sensor. On NTU-RGB+D the cameras are fixed, while on Kinetics the videos are usually shot by hand-held devices, leading to large camera motion. The fact that the proposed ST-GCN can work well on both datasets demonstrates the effectiveness of the proposed spatial temporal graph convolution operation and the resultant ST-GCN model.

We also notice that on Kinetics the accuracies of skeleton based methods are inferior to video frame based models (?). We argue that this is due to a lot of action classes in Kinetics requires recognizing the objects and scenes that the actors are interacting with. To verify this, we select a subset of 30 classes strongly related with body motions, named as “Kinetics-Motion” and list the mean class accuracies of skeleton and frame based models (?) on this subset in Table 4. We can see that on this subset the performance gap is much smaller. We also explore using ST-GCN to capture motion information in two-stream style action recognition. As shown as in Fig. 5, our skeleton based model ST-GCN can also provide complementary information to RGB and optical flow models. We train the standard TSN (?) models from scratches on Kinetics with RGB and optical flow models. Adding ST-GCN to the RGB model leads to 0.9% increase, even better than optical flows (0.8%). Combining RGB, optical flow, and ST-GCN further raises the performance to 71.7%. These results clearly show that the skeletons can provide complementary information when leveraged effectively (e.g. using ST-GCN).

	RGB TSN	Flow TSN	ST-GCN	Acc(%)
Single Model	✓			70.3
		✓		51.0
			✓	30.7
Ensemble Model	✓	✓		71.1
	✓		✓	71.2
	✓	✓	✓	71.7

Table 5: Class accuracies on the Kinects dataset **without** ImageNet pretraining. Although our skeleton based model ST-GCN can not achieve the accuracy of the state of the art model performed on RGB and optical flow modalities, it can provide stronger complementary information than optical flow based model.

5 Conclusion

In this paper, we present a novel model for skeleton based action recognition, the spatial temporal graph convolutional networks (ST-GCN). The model constructs a set of spatial temporal graph convolutions on the skeleton sequences. On two challenging large-scale datasets, the proposed ST-GCN outperforms the previous state-of-the-art skeleton based model. In addition, ST-GCN can capture motion information in dynamic skeleton sequences which is complementary to RGB modality. The combination of skeleton based model and frame based model further improves the performance in action recognition. The flexibility of ST-GCN model also opens up many possible directions for future works. For example, how to incorporate contextual information, such as scenes, objects, and interactions into ST-GCN becomes a natural question.

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